

CE 526 Finite Element Method

Assignment # 6 (due June 22nd, 2026)

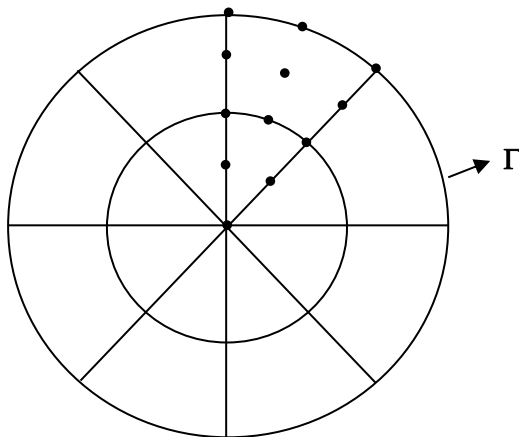
IMPORTANT NOTE: I WILL NOT BE ABLE TO ACCEPT ANY LATE SUBMISSIONS THIS TIME SINCE I SHOULD GRADE YOUR FINAL ASSIGNMENT WITHIN A VERY TIGHT SCHEDULE BEFORE THE GRADE SUBMISSION DEADLINE. PLEASE SUBMIT WHATEVER YOU CAN.

You are given the following circular membrane problem in 2D:

$$\begin{aligned} -T \nabla^2 u &= p(x, y) \text{ in } \Omega \\ u &= 0 \text{ on } \Gamma \end{aligned}$$

where Ω is the circular region shown below subject to the Dirichlet boundary condition Γ on the boundary. T is the constant tension in the membrane and $p(x, y)$ is the uniformly- distributed transverse load on the membrane.

Consider the case where $p(x, y)=1$ and $T=1$. Assume that the radius of the membrane is 2.



Taking advantage of the symmetry, and using only the given single isoparametric biquadratic quadrilateral element and the isoparametric quadratic triangular element shown on the membrane:

- Determine the solution at the nodes
- Plot the solution u on a radial line (please note that this is a 2D plot, not 3D so it saves us time!)
- Discuss your results in detail (also include how you may improve the accuracy of this result)

Hint: You can place the origin $(x, y)=(0, 0)$ at the center of the circle.